GlobalTrade Logistics Corp

Supply Chain Operations & Data Management

Internal Business Description Document

Document Owner	Supply Chain Data Governance
Classification	Confidential — Internal Use Only
Version	2.0
Last Updated	March 2025
Review Cycle	Annual
Next Review	March 2026

Table of Contents

1. Company Overview
2. Supplier Relationships and Procurement
3. Warehouse Management and Inventory
4. Inbound Logistics: Shipments and Delivery Events
5. Quality Control
6. Sales and Outbound Distribution
7. Returns Management
8. Data Integration and Analytical Use Cases
9. Data Model Summary
10. Key Performance Indicators
11. Appendix: Entity Relationship Overview

1. Company Overview

GlobalTrade Logistics Corp (GTL) is a multinational manufacturing and distribution company operating across **six continents**. Founded in 1995, the company has grown from a regional parts

distributor into an integrated supply chain operator managing the full lifecycle of products — from raw material sourcing through to end-customer delivery.

> ***Our Mission:*** *To deliver operational excellence across the global supply chain through data-driven decision-making, supplier collaboration, and customer-centric logistics.*

Business Segments

Segment	Description
Industrial Components	Precision-engineered parts for manufacturing
Consumer Goods	Retail-ready products for direct distribution
Specialty Chemicals	Temperature-controlled and hazardous materials

Company at a Glance

Metric	Value
Founded	1995
Headquarters	Rotterdam, Netherlands
Active Suppliers	60+ worldwide
Active Customers	~150 (retail, wholesale, OEM, e-commerce)
Distribution Centers	10 global locations
Product SKUs	100+ across 12 categories
Total Warehouse Capacity	250,000+ m^2
Quarterly Purchase Orders	~400
Quarterly Sales Orders	~500

---

2. Supplier Relationships and Procurement

Supplier management is a cornerstone of GTL's operations. We maintain a curated base of **60 approved vendors**, each evaluated on quality, reliability, pricing, and geographic coverage.

Supplier Evaluation Framework

Criterion	Weight	Measurement
On-time Delivery	35%	% of POs delivered within expected window
Quality Score	30%	Inspection pass rate and defect counts
Pricing Competitiveness	20%	Benchmarked against market rates
Responsiveness	15%	Average response time to inquiries

Each supplier receives a **performance rating** (3.0 to 5.0) that is refreshed quarterly and used to prioritize order allocation.

Product Categories

Suppliers are categorized across 12 product families:

Category	Subcategories
Electronics	Semiconductors
Raw Materials	—
Packaging	—
Mechanical Parts	Automotive
Chemicals	—
Textiles	—
Consumer Goods	—
Industrial Equipment	—
Food & Beverage	—
Pharmaceuticals	—

Purchase Order Workflow

Purchase orders (POs) are the primary mechanism for replenishment:

```
Draft → Submitted → Approved → In Transit → Received
                                  └──→ Cancelled
```

PO Metric	Quarterly Volume
Purchase Orders	~400
Order Lines (avg. 3/PO)	~1,200
Order Value Range	$500 – $150,000

Each PO references a single supplier and includes order date, expected delivery date, total amount, and detailed line items (product, quantity, unit price, line total).

---

3. Warehouse Management and Inventory

Warehouse Network

GTL operates 10 strategically located facilities:

Location	Type	Function
Shanghai	Fulfillment Center	Asia-Pacific direct-to-consumer
Rotterdam	Distribution Center	European bulk replenishment
Singapore	Transit Hub	Cross-docking and transshipment
Dubai	Regional DC	Middle East and Africa distribution
Los Angeles	Distribution Center	Americas West Coast hub
Hamburg	Cold Storage	Temperature-sensitive products
Tokyo	Fulfillment Center	Japan direct distribution
Mumbai	Regional DC	Indian subcontinent operations
Antwerp	Transit Hub	European cross-docking
Busan	Distribution Center	Korean peninsula and East Asia

Inventory Management

Inventory is tracked at the **product × warehouse** level. Each position records:

- **Quantity on hand** — Current stock level

- [redacted] — Threshold for automatic replenishment
- **Last restock date** — Supports cycle counting and aging analysis

[redacted]

Inventory Metric	Value
Active Inventory Positions	~500
Products Tracked	100 SKUs
[redacted]	
Warehouses	10

When quantity on hand falls at or below the reorder level, the system triggers replenishment workflows automatically.

---

4. Inbound Logistics: Shipments and Delivery Events

[redacted]

When a purchase order is fulfilled by a supplier, goods are shipped to one of our warehouses.

Carrier Partners

Carrier	Primary Routes
Maersk	Ocean freight — Asia to Europe
CMA CGM	Ocean freight — Asia to Americas
MSC	Ocean freight — Global
DHL	Air and express — Global
FedEx	Air and express — Americas
UPS	Ground and express — Americas
Kuehne+Nagel	Multimodal — Global
DB Schenker	European road freight
XPO Logistics	North American logistics
CEVA	Contract logistics

Shipment Lifecycle

Status	Description
Pending	Awaiting pickup from supplier
Picked Up	Collected by carrier
In Transit	En route to destination warehouse
Delivered	Received at destination
Delayed	Behind schedule
Exception	Issue requiring intervention

We process approximately **600 shipments per quarter**.

Delivery Event Tracking

Delivery events provide granular visibility into shipment progress:

Event Type	Description
Pickup	Goods collected from supplier
Departure	Left origin facility
Arrival	Reached intermediate or final facility
Customs Clearance	Cleared border controls
Delivery	Final delivery confirmed
Exception	Damage, loss, or routing issue
Delay	Schedule deviation recorded

We record approximately **1,500 delivery events per quarter**, enabling real-time tracking and exception management.

---

5. Quality Control

Quality inspections are performed on incoming shipments before goods enter inventory.

Inspection Process

Each inspection records:

- [redacted] — What was inspected
- **Inspector ID** — Who performed the inspection
- [redacted], or Pending Review
- **Defect count** — Number of defective units
- **Notes** — Observations and recommended actions

QC Metric	Quarterly Volume
Inspections Performed	~300
Average Pass Rate	Tracked per supplier
Defect Resolution Time	< 5 business days

Failed or conditional results trigger:

1. Quarantine of affected goods
2. Supplier notification and corrective action request
3. Scorecard adjustment

---

6. Sales and Outbound Distribution

Customer Base

GTL serves **150 active customers** across multiple segments:

Customer Type	Description
Retail	Brick-and-mortar stores and chains
Wholesale	Bulk buyers and distributors
Distributor	Regional and specialty distributors
Manufacturer	OEM and contract manufacturers
E-commerce	Online retailers and marketplaces

Customer **credit limits** are maintained to support order approval workflows.

Sales Order Workflow

```
Draft → Confirmed → Picking → Shipped → Delivered
```

```
                                    └──→ Cancelled
```

Sales Metric	Quarterly Volume
Sales Orders	~500
Order Lines (avg. 3/order)	~1,500
Order Value Range	$100 – $75,000

Each sales order references a customer and a fulfilling warehouse. Ship date is recorded when the order leaves the facility.

---

7. Returns Management

Return requests are initiated by customers and linked to the original sales order.

Return Reasons

Reason	Description
Defective	Product does not function as specified
Wrong Item	Incorrect product shipped
Damaged in Transit	Shipping damage
Customer Change	Buyer changed requirements
Quality Issue	Product does not meet expectations
Expired	Product past its use-by date

Return Workflow

```
Requested → Approved → Received → Refunded
                 └──→ Rejected
```

Returns Metric	Quarterly Volume
Return Requests	~200

Average Refund Amount	Varies by product

Returns data supports reverse logistics planning, supplier quality feedback, and customer satisfaction analysis.

---

8. Data Integration and Analytical Use Cases

The supply chain data model supports end-to-end visibility from supplier to customer. Key analytical capabilities include:

Order Fulfillment Tracking

```
Purchase Order → Shipment → Delivery Events → Warehouse Receipt → Inventory Updat
```

Core Analytics

Use Case	Data Sources	Business Value
Inventory Optimization	Inventory, Purchase Orders, Sales Orders	Reduce stockouts and excess inventory
Supplier Performance	Suppliers, POs, Shipments, Inspections	Optimize vendor selection and negotiation
Customer Analytics	Customers, Sales Orders, Returns	Improve retention and order accuracy
Logistics Efficiency	Shipments, Delivery Events, Carriers	Reduce transit times and exception rates
Quality Trending	Inspections, Suppliers, Products	Early detection of quality deterioration

---

9. Data Model Summary

Master Data Entities

Entity	Description	Approx. Records

Supplier	Approved vendors	60
Product Category	Product classification hierarchy	12
Product	SKU-level product catalog	100
Warehouse	Distribution and fulfillment centers	10
Customer	Buyers and trading partners	150

Transactional Entities

Entity	Description	Approx. Records
Inventory	Stock positions (product × warehouse)	500
Purchase Order	Supplier replenishment orders	400
Order Line	PO line-item detail	1,200
Shipment	Inbound goods movement	600
Delivery Event	Shipment tracking milestones	1,500
Sales Order	Customer orders	500
Sales Order Line	SO line-item detail	1,500
Quality Inspection	Incoming goods inspection	300
Return Request	Customer returns	200

Total records: ~7,032

10. Key Performance Indicators

KPI	Target	Measurement
Order Fill Rate	≥ 95%	% of orders shipped complete
On-time Delivery (Inbound)	≥ 90%	% of POs received by expected date
On-time Delivery (Outbound)	≥ 95%	% of SOs delivered by promised date
Inventory Turnover	≥ 8x/year	COGS / Average Inventory